

LESSON 8.1

SOLVING EQUATIONS



LESSON INTRODUCTION

MA.8.AR.2.1 Solve multi-step linear equations in one variable, with rational number coefficients. Include equations with variables on both sides.

LEARNING TARGETS

- I can solve multi-step linear equations from a mathematical context.

BENCHMARK COHERENCE

BUILDING ON

MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical or real-world context, where all terms are rational numbers.

WORKING TOWARD

MA.912.AR.1.2 Rearrange equations or formulas to isolate a quantity of interest.

MA.912.AR.2.1 Given a real-world context, write and solve one-variable multi-step linear equations.

ESSENTIAL QUESTIONS

- How do you solve an equation in one variable?

LESSON NARRATIVE

This lesson begins by reviewing concepts learned in prior grades when solving two-step equations before expanding into solving multi-step equations. Students will revisit applying the distributive property as well as combining like terms. The lesson focuses on students exploring multiple approaches to solving an equation and recognizing that there is not a single “correct” set of steps to use. As students work with different equations throughout the unit, it is critical they develop flexibility in solving equations in different ways and are given opportunities to discuss and reflect on their strategies.

COMPONENT SUMMARY

8.1.1 WARM-UP (~5 MIN.)

Problem-solving: video and response

8.1.2 COLLABORATION (10-15 MIN.)

Consider different strategies to solve equations

8.1.3 GUIDED INSTRUCTION (5 MIN.)

Review terms and like terms

8.1.4 COLLABORATION (5-10 MIN.)

Error analysis when combining like terms

8.1.5 GUIDED PRACTICE (10-15 MIN.)

Solve multi-step equations with the variable on one side

8.1.6 WRAP-UP (~5 MIN.)

Explain an error in solving an equation

RESOURCES

GLOSSARY

Key Terms:

- Term
- Like terms

Additional Terms: N/A

MATERIALS

- Problem-solving video (Warm-Up)
- (Optional) Colored pencils
- (Optional) Chart paper

ADDITIONAL PREPARATION

- Prepare the problem-solving video for the Warm-Up.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may try to combine unlike terms such as rewriting $3a+18$ as 21 or $21a$.
- When distributing a number, students may distribute to the first term but forget the second or third.
- Students may disregard the negative sign in front of a term or a number being distributed.
- Students may struggle to recognize different ways of approaching how to solve an equation.

THINGS TO CONSIDER

- Consider the different strategies that could be used to solve the equations in the lesson so that they can be more easily recognized when observing student thinking. This allows for the lesson to be built on student understanding.

LITERACY AND LANGUAGE ROUTINES

Algebra Talk

The lesson offers the opportunity for students to explore different approaches to solving equations. After displaying an equation, give students a few moments to think of a strategy they would use to get started. Encourage students to share their different strategies. Then discuss the benefits of each for various equations.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.4.1 Discussion and Reflection

The 8.1.4 Collaboration and 8.1.6 Wrap-Up provide the opportunity for students to engage in an error analysis of solving an equation. Use these activities to provide students with the opportunity to examine their own thinking and to learn from the thought processes of other students. Encourage students to take advantage of the discussion opportunities throughout to practice explaining their own thinking and reasoning.

DIFFERENTIATE INSTRUCTION

Consider having students highlight or circle like terms in the same color before solving an equation. For students who are visually impaired or who may feel overwhelmed by a string of numbers and variables, consider rewriting the equations on chart paper for students to solve there. This may be especially helpful when several problems appear beside each other in vertical rows such as in the 8.1.5 Guided Practice. Provide opportunities for students to verbally explain the process of solving the equations.

8.1.1 WARM-UP: PROBLEM SOLVING

QUESTIONING

Encourage students to think of specific careers when collaboration would be particularly helpful. Allow students to research one of those careers and share information about it with the class as a homework or extra credit opportunity.



8.1.1 Warm-Up: Problem Solving

Employers seek out individuals who can collaborate with others to solve problems.

1. What are some of the advantages of working with others when problem solving?

Answers will vary. Working together provides multiple viewpoints to draw on different people's strengths. Also, one person may have an insight that another person misses.

2. Think of a situation outside of school when you exchanged ideas with someone else to solve a problem. Summarize the steps you took to solve the problem and arrive at the final solution.

Answers will vary. Some students may reflect on projects at home, such as a repair, in which they identified the problem, worked with another family member to plan how to fix it, and then worked together to repair it. Other examples may include planning an event or special meal.



8.1.2 Collaboration: Solving Equations

- Da'Liyah and Gerald both started to solve the equation $3(x - 7) = 18$ for x . Their first steps are shown in green, followed by the resulting equation.

Da'Liyah	Gerald
$\frac{3(x - 7)}{3} = \frac{18}{3}$ $x - 7 = 6$	$3(x - 7) = 18$ $3(x) - 3(7) = 18$ $3x - 21 = 18$
$x - 7 + 7 = 6 + 7$ $x = 13$	$3x - 21 + 21 = 18 + 21$ $\frac{3x}{3} = \frac{39}{3}$ $x = 13$

- Discuss with your partner what Da'Liyah did differently from Gerald.
Da'Liyah divided both sides of the equation by 3, whereas Gerald used the distributive property initially.
- Finish solving the equation for both Da'Liyah and Gerald, continuing from their first steps.
- Verify your answers with your partner and if necessary, resolve any differences.
- What do you notice about Da'Liyah's and Gerald's solutions to the equation?
The solutions are the same.

- Two equations are shown in the table.

- For each equation, solve it using either Da'Liyah's or Gerald's strategy.

Equation	Da'Liyah's Strategy	Gerald's Strategy
$-3 = -10(b - 1)$	$\frac{-3}{-10} = \frac{-10(b - 1)}{-10}$ $\frac{3}{10} = b - 1$ $\frac{3}{10} + 1 = b$ $b = 1\frac{3}{10}$	$-3 = -10(b - 1)$ $-3 = -10(b) - (-10)(-1)$ $-3 = -10b + 10$ $-3 - 10 = -10b$ $-13 = -10b$ $\frac{-13}{-10} = \frac{-10b}{-10}$ $b = 1\frac{3}{10}$
$2000(k - 2) = 6000$	$\frac{2000(k - 2)}{2000} = \frac{6000}{2000}$ $k - 2 = 3$ $k = 3 + 2$ $k = 5$	$2000(k - 2) = 6000$ $2000(k) - 2000(2) = 6000$ $2000k - 4000 = 6000$ $2000k = 6000 + 4000$ $2000k = 10,000$ $\frac{2000k}{2000} = \frac{10,000}{2000}$ $k = 5$

- Discuss with your partner why they chose to utilize the specific strategy for each equation. Summarize their reasoning.
Answers will vary. Dividing first may allow you to arrive at smaller values sooner. Applying the distributive property removes parentheses which makes the equation easier to manipulate.

8.1.2 COLLABORATION: SOLVING EQUATIONS

TEACHER MOVES

This is a discussion opportunity. Students do not need to write an explanation here, but a sample response is provided for reference.

Listen as students discuss their thinking in response to this prompt to determine student thinking to highlight or misconceptions to address.

TEACHER MOVES

There is not a “correct” way to solve either equation. Students may choose different strategies for each equation or stick with the same one. Discussing and experimenting with different strategies helps build understanding, flexibility, and fluency when solving equations.

This component allows the opportunity for the teacher to observe students’ comfort with different strategies for solving equations prior to 8.1.3 Guided Instruction. Common misconceptions could be addressed in a whole-group debrief as a segue into the next component.

8.1.3 GUIDED INSTRUCTION: LIKE TERMS

DIFFERENTIATE INSTRUCTION

To help students identify like terms, recommend students to highlight or circle like terms with the same color before they begin solving an equation. Identifying the like terms before beginning the algebraic process reduces the details that students must consider when solving the equation.



8.1.3 Guided Instruction: Like Terms

A **term** is a single number or variable, or numbers and variables multiplied together. For example, the expression $0.5 - 4y$ has two terms.

The equation below includes four terms.

$$a - 21 + 3a = 18$$

- a is a term with a single variable.
- -21 and 18 are each terms with a single number, often referred to as constant terms.
- $3a$ is a term with a number and a variable multiplied together.

When solving equations, only **like terms** can be added or subtracted together. In the equation above, a and $3a$ are like terms and -21 and 18 are like terms.

Like terms are terms with the exact same variable(s) raised to the same exponent.

1. Draw lines to match the like terms in each column.

A. $10p$	I. -1
B. $12m^3$	II. $13m^3$
C. f	III. $-3p$
D. -64	IV. $7f$

(Red lines connect A to IV, B to II, C to III, and D to I.)

8.1.4 COLLABORATION: ERROR ANALYSIS

TEACHER MOVES

Students need to understand that like terms can be combined on either side of the equation or when performing an operation on both sides of the equation (i.e. subtracting 6 from both sides), but like terms cannot simply be combined from one side of the equal sign to the other.

TEACHER MOVES

This is a discussion opportunity. Students do not need to write an explanation here unless directed to do so by the teacher, but a sample response is provided for reference.

Listen as students discuss their thinking in response to this prompt to determine student thinking to highlight or misconceptions to address.



8.1.4 Collaboration: Error Analysis

Jayla was given the equation $15 = 3c + 6 + 6c$ to solve.

She combined the terms $3c$ and $6c$ by adding them together to create the equation $15 = 9c + 6$.

1. Complete the statement.

Jayla ☒ is ☐ is not correct to combine the terms $3c$ and $6c$ because they are ☒ like terms. ☐ not like terms.

Given the same equation, Kenneth combined the terms $3c$, 6 , and $6c$ by adding them together to create the equation $15 = 15c$.

2. Complete the statement.

Kenneth is ☐ correct ☒ incorrect to combine the terms $3c$, 6 , and $6c$ because they are ☐ like terms. ☒ not like terms.

3. Discuss with your partner whether or not the equation can be rewritten by adding 15 and 6 together.
No, the equation cannot be rewritten by adding 15 and 6 together because even though 15 and 6 are like terms, they cannot be combined because they are on different sides of the equal sign.



8.1.5 Guided Practice: Solving Equations

1. Consider the equation $12 = 4(1 + q - 2)$.
 - a. Explain the first step you would take to solve the equation for q .
Answers will vary. Combine 1 and -2; Use the distributive property to multiply 4 by the terms in the parentheses; Divide both sides of the equation by 4
 - b. Compare your first step with your partner.
 If you chose different first steps, explain why both are correct. If you chose the same first step, work together to describe another first step that could have been taken.
Answers will vary based on responses to part a.
 - c. Solve the equation for q .
 $q = 4$
 - d. Use substitution to verify the solution.
 $12 = 4(1 + 4 - 2)$
 $12 = 4(3)$
 $12 = 12$

8.1.5 GUIDED PRACTICE: SOLVING EQUATIONS

TEACHER MOVES

Algebra Talk

8.1.5 Guided Practice offers students the opportunity to explore different approaches to solving equations. After displaying an equation, give students a few moments to think of a strategy they would use. Encourage students to share their different strategies. Then discuss the benefits of each approach.

8.1.5 GUIDED PRACTICE: SOLVING EQUATIONS (CONTINUED)

DIFFERENTIATE INSTRUCTION

Use observational data from the 8.1.2 and 8.1.4 Collaborations to determine how much guidance is needed to address key ideas in this Guided Practice component. If necessary, consider modeling question 2 or pulling a small group as they work through that problem to address common misconceptions shown in the Collaborations.

TEACHER MOVES

Pay close attention to the problem $13=g+2(g-4)$. Students may mistakenly try to divide by 2 as the first step. If observed, discuss that this is possible only if all terms are divided by 2, not just 13 and $2(g-4)$. Consider showing the result of dividing both sides by 2 in this case to ensure understanding: $7.5 = \frac{g}{2} + g - 4$. It's not that this is not a strategy that could be used, but it does need to be used properly.

DIFFERENTIATE INSTRUCTION

For students who are visually impaired or who may feel overwhelmed by a lot of numbers and variables, consider rewriting the problems on chart paper for students to solve there. This will allow students to focus on each individual equation and may particularly benefit students with dysgraphia or dyscalculia.

2. Three equations are given.

- a. Solve each equation on your own using the space provided to show your work.

Solution methods may vary.

$12 = 12(-2 - z + 5)$	$2w + 9 - 5w = -3$	$13 = g + 2(g - 4)$
$1 = -2 - z + 5$ $1 = 3 - z$ $-2 = -z$ $2 = z$	$-3w + 9 = -3$ $-3w = -12$ $w = 4$	$13 = g + 2g - 8$ $13 = 3g - 8$ $21 = 3g$ $7 = g$

- b. Compare your work and solutions with your partner. Resolve any differences in solutions, if necessary.

Students may use different approaches to solve the problems, but should arrive at the same solutions regardless.



8.1.6 Wrap-Up: Error Analysis

1. Hubert solved the equation $-8 = 4(2 - h + 1) + 4 - 4h$.

His work is shown.

$$\begin{aligned} \frac{-8}{4} &= \frac{4(2-h+1)}{4} + 4 - 4h \\ -2 &= 2 - h + 1 + 4 - 4h \\ -2 &= 7 - 5h \\ -9 &= -5h \\ 1.8 &= h \end{aligned}$$

- a. Explain the mistake Hubert made when solving the equation.
When Hubert divided by 4 to start, he only divided part of the right side of the equation when he should have divided all terms on the right side by 4.
- b. Write a short note to Hubert with a tip for how to avoid this error in the future so he can learn from his mistake.
Hubert, if you are going to begin solving an equation by dividing both sides by a value, be sure to divide all of the terms in the equation by that value. Otherwise, the expressions on either side will no longer be equivalent.

8.1.6 WRAP-UP: ERROR ANALYSIS

DIFFERENTIATE INSTRUCTION

The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.

DIFFERENTIATE INSTRUCTION

Some students may benefit from visually pointing out the error and/or verbally explaining the mistake rather than responding in writing. Allow students to complete the activity in whatever mode is most comfortable for them.